

Aakash Shahani

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EDUCATION

University of South Florida (USF)

Graduation: December 2025

Bachelor of Science in Computer Science

GPA: 3.5/4.0

- Relevant coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Database Design, Big Data Storage & Analysis, Probability & Statistics, Software Engineering

TECHNICAL SKILLS

Languages: Python, Go, C/C++, Java, TypeScript, JavaScript, SQL, Bash

AI / ML: PyTorch, transformers, Hugging Face, scikit-learn, XGBoost, RAG, Embeddings, LangChain, Prompt Engineering, LLM APIs (Anthropic/OpenAI), SHAP

Data & Cloud: Airflow, dbt, Iceberg, Spark, Kafka, star schema & SCD2, Great Expectations, AWS (S3/Glue/Athena), Terraform, Docker, CI/CD (GitHub Actions), pandas, NumPy, A/B Testing, Prometheus/Grafana

Backend, Databases & Tools: FastAPI, REST, GraphQL, Node.js, Next.js, React, PostgreSQL, MySQL, MongoDB, Redis, pgvector/FAISS, Metabase, Git, Linux, pytest

EXPERIENCE

Machine Learning Research Assistant

Oct 2025 – Present

CSSAI Lab, University of South Florida

Tampa, FL

- Benchmarked **3** model families (fine-tuned RoBERTa, a GPT-class model via the OpenAI Batch API, and few-shot LLaMA 3 on Groq) for turn-level negotiation-behavior classification, validated against a human-labeled gold set at Cohen's κ **0.76** across **1,030** turns.
- Built a reproducible batch-inference and evaluation harness (versioned JSONL datasets, automated scoring) enabling one-command model comparisons and ablations over the full labeled corpus.

Deep Learning Research Assistant

Aug 2024 – Apr 2025

University of South Florida

Tampa, FL

- Engineered an end-to-end ETL pipeline pulling protein sequences from the UniProt REST API and applying MMseqs2 clustering at a **30%** similarity threshold to deduplicate and standardize a model-ready dataset.
- Designed and trained an LSTM binding-site classifier (BCE loss, Adam with grid search, dropout, LR scheduling), benchmarking against CLAPE-SMB and SCRIBER with k-fold cross-validation and a held-out test set.

PROJECTS

ScholarLens: Agentic Research Platform | *FastAPI, Next.js, pgvector, Claude API*

[GitHub](#) | [Live Demo](#)

- Built an agentic RAG platform that flags cross-paper contradictions via a two-stage BM25 + dense pre-filter and an LLM judge at macro-F1 **0.788** / κ **0.683**, cutting **~4,500** LLM calls by over **90%**.
- Ran **6** analyses per paper concurrently for a **5×** speedup and served pgvector (HNSW) search behind a cross-encoder reranker at nDCG@5 **0.98**; deployed live with multi-user isolation.

TradePulse: Real-Time Streaming Pipeline | *Kafka, Spark Streaming, Docker*

[GitHub](#) | [View Project](#)

- Ingested live Coinbase trades through a **6**-partition Kafka topic into 1-minute OHLC candles in Spark Structured Streaming; guaranteed exactly-once output via watermarks and idempotent writes, proven by a mid-stream crash test with **0** duplicate candles.
- Sustained sub-second **600–800ms** micro-batches under **~1,500** rows/sec bursts across **9** Dockerized services with Prometheus/Grafana observability; validated by **20** tests in CI.

NewsRec: Cold-Start News Recommender | *PyTorch, FAISS, FastAPI, Docker*

[GitHub](#) | [View Project](#)

- Trained a content-based two-tower retriever on the MIND dataset that beat popularity/SVD/ALS baselines by **+0.124** AUC (**~23%**) across **73,152** impressions by embedding article text to solve cold-start.
- Served candidate generation at **2.8ms** p99 over **3,000** users via a FAISS HNSW index so the FastAPI path never loads a model at request time; Docker + **17** tests in CI.

UrbanFlow: Data Lakehouse Platform | *Airflow, dbt, Apache Iceberg, AWS, Terraform*

[GitHub](#) | [View Project](#)

- Built an end-to-end data lakehouse over **3M** NYC taxi trips: Airflow into Apache Iceberg on S3, dbt-modeled into a star schema (fact + SCD Type 2 dims), deployed on AWS with Terraform and queried on Athena/Trino.
- Cut Athena bytes scanned **97%** (**4.81→0.13MB**) via day-partitioning; gated it with Great Expectations and **17** dbt tests that block bad data before the gold layer; idempotent Iceberg MERGE held **2.87M** rows at **0** duplicates.